

Quick fix: On India's Research Development and Innovation scheme

Budgetary allowances alone won't solve India's R&D problem

The Union Cabinet recently approved a ₹1-lakh crore Research Development and Innovation (RDI) scheme that aims to **incentivise** the private sector to invest in basic research. The scheme will primarily consist of a special purpose fund **established** within the Anusandhan National Research Foundation (ANRF), which will act as the **custodian** of funds. The funds will be in the form of low-interest loans. The ANRF is **conceived** as an independent institutional body, with **oversight** by the Science Ministry, to **allocate** funds for basic research and to incentivise private sector participation in core research. The **involvement** of the ANRF here **is** a **novel** move as the newly created organisation is meant to be the equivalent of a **single-window clearance mechanism** for funding research and development for universities and academic institutions. It is also expected to get about 70% of its budget from private sources. **In sum**, through the RDI and the ANRF, the government is looking to **stake the bold claim** that it has **played its part** and that it is now up to the private sector to come forward and reverse the ratio from where the government today **accounts for** about 70% of India's R&D spend. However, already **incipient** in the government's **tall ambitions** are **traces** of what has caused previous such schemes to **falter**. The first of these is **conservatism**.

It **turns out** that a condition for **availing** funds is that only products that have reached a certain level of development and market **potential** or, what are called Technology Readiness Level-4 (TRL-4) projects, would be eligible. There are nine TRL levels, a **hierarchy** that was first conceived by the United States' National Aeronautics and Space Administration (NASA) in the 1970s. TRL-1 represents a basic level of research and TRL-9 a **state** of advanced **readiness**. TRL-4 appears to be an **arbitrary** decision to support any **promising** research that has progressed halfway. Were there such **a magic sauce**, **venture capital** industries, **premised on** the **fickleness** of predicting the 'next big thing', **would** not exist. The scheme also seems to forget that technologically advanced countries have become what they are because of their military industrial **complexes** — where the **spectre** of war incentivises the development of technology that is risky and expensive but, over time, may prove to be of **immense** civilian value — examples are the Internet or the Global Positioning System. India continues to **haemorrhage** scientists to the West due to the lack of opportunities **commensurate** with their training. Finally, it lacks a deeply skilled manufacturing sector that can make the products that scientists conceive **of**. Budgetary allowances cannot **overnight** fix that which requires major surgery.

- Red/blue coloring of words in the sentence indicates subject verb relationship; where 'red' denotes 'subject' and 'blue' denotes 'verb'.

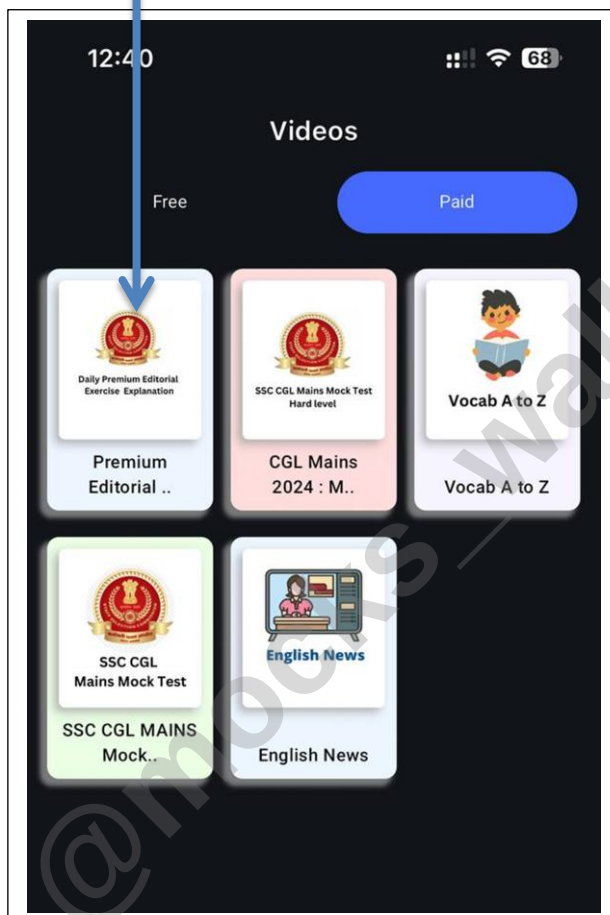
[Practice Exercise]

Vocabulary

1. **Quick fix** (phrase) – an easy remedy or solution, especially a temporary one which fails to address underlying problems.
तात्कालिक समाधान
2. **Budgetary** (adjective) – financial, fiscal, economic, monetary, funding-related
बजटीय
3. **Incentivise** (verb) – encourage, motivate, stimulate, promote, induce प्रोत्साहित करना
4. **Establish** (verb) – set up, found, create, initiate, institute स्थापित करना
5. **Custodian** (noun) – guardian, keeper, caretaker, overseer, steward संरक्षक
6. **Conceive** (verb) – imagine, formulate, devise, invent, originate कल्पना करना / बनाना
7. **Oversight** (noun) – supervision, control, regulation, governance, surveillance निगरानी
8. **Allocate** (verb) – distribute, assign, allot, apportion, designate आवंटित करना
9. **Novel** (adjective) – new, innovative, original, fresh, unprecedented नया / नवीन
10. **Single-window clearance** (noun) – a centralized online portal or platform that streamlines the process of obtaining various business approvals and clearances
11. **Mechanism** (noun) – system, process, structure, method, procedure प्रणाली / तंत्र
12. **In sum** (phrase) – in conclusion, overall, to summarize, briefly, in short संक्षेप में
13. **Stake the claim** (phrase) – assert, declare, proclaim, affirm, lay claim दावा करना
14. **Bold** (adjective) – daring, courageous, brave, audacious, confident साहसी / दुस्साहसी
15. **Play one's part** (phrase) – contribute, participate, cooperate, fulfill role अपना योगदान देना
16. **Account for** (phrasal verb) – constitute, make up, explain, represent, clarify हिस्सा होना
17. **Incipient** (adjective) – beginning, initial, nascent, emerging, budding आरंभिक / प्रारंभिक अवस्था में
18. **Tall ambition** (phrase) – a very ambitious goal or aspiration, one that is high and challenging to achieve बड़ी महत्वाकांक्षा
19. **Trace** (verb) – detect, find, identify, track, discover पता लगाना / संकेत मिलना

20. **Falter** (verb) – weaken, hesitate, stumble, fail, decline विफल होना / लड़खड़ाना
21. **Conservatism** (noun) – caution, traditionalism, moderation, restraint, orthodoxy परंपरागत सोच
22. **Avail** (verb) – make use of, utilize, benefit from, take advantage of लाभ उठाना
23. **Potential** (noun) – possibility, capability, capacity, promise, likelihood क्षमता
24. **Hierarchy** (noun) – ranking, order, grading, structure, pecking order श्रेणी
25. **State** (noun) – condition, level, stage, situation, phase अवस्था
26. **Readiness** (noun) – preparedness, willingness, alertness, promptness, availability तत्परता
27. **Arbitrary** (adjective) – random, capricious, subjective, whimsical, unreasoned मनमाना
28. **Promising** (adjective) – hopeful, encouraging, optimistic, bright, prospective आशाजनक
29. **A magic sauce** (noun) – secret formula, perfect solution, miracle fix, magic trick, wonder recipe जादुई उपाय
30. **Venture capital** (noun) – a form of private equity financing provided by firms or funds to startup, early-stage, and emerging companies उद्यम पूंजी
31. **Premise** (on) (verb) – base on, found on, establish on, ground in आधारित होना
32. **Fickleness** (noun) – unpredictability, inconsistency, instability, unreliability, changeability अस्थिरता / चंचलता
33. **Complex** (noun) – system, structure, network, setup, institution जटिल प्रणाली
34. **Spectre** (noun) – fear, threat, ghost, shadow, danger डर / आशंका
35. **Immense** (adjective) – huge, vast, enormous, massive, tremendous बहुत बड़ा
36. **Haemorrhage** (verb) – lose rapidly, drain, bleed, leak, suffer heavy loss बड़ी संख्या में खो देना
37. **Commensurate** (adjective) – proportionate, corresponding, equivalent, matching, appropriate अनुरूप / समानुपाती
38. **Overnight** (adjective) – sudden, immediate, instant, unplanned, hasty रातोंरात / अचानक

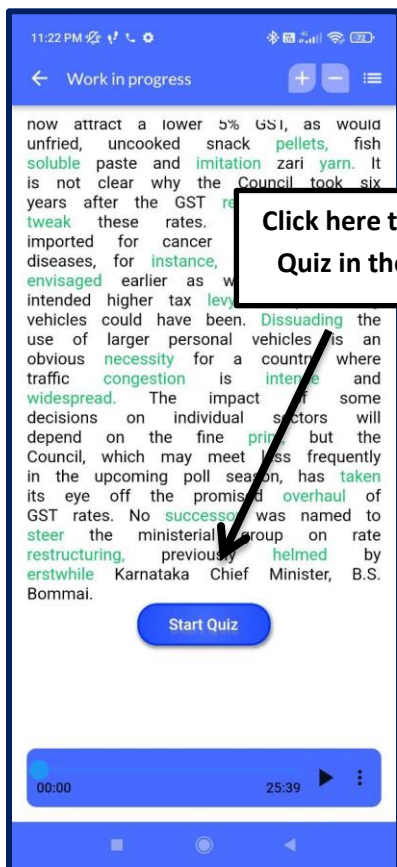
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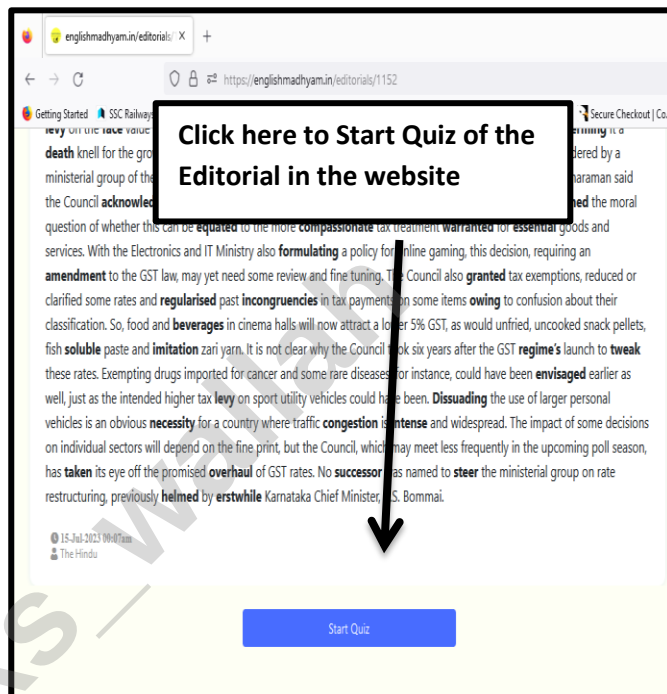
Summary of the Editorial

1. The Union Cabinet has approved a ₹1-lakh crore Research Development and Innovation (RDI) scheme to encourage private sector investment in basic research.
2. A special purpose fund within the Anusandhan National Research Foundation (ANRF) will manage and distribute these funds.
3. The funds will be disbursed as low-interest loans to eligible projects, not as direct grants.
4. The ANRF is intended to be a centralized, single-window body for research funding, under the oversight of the Science Ministry.
5. The government expects 70% of ANRF's funding to come from private sources, showing a clear push toward private participation.
6. With RDI and ANRF, the government is signaling that it has fulfilled its role and now expects the private sector to lead R&D spending.
7. Currently, the government bears about 70% of India's R&D spending, a ratio it wants to reverse.
8. However, the scheme shows signs of policy conservatism, which has led to the failure of similar past efforts.
9. A key limitation is that funding will be limited to projects at Technology Readiness Level-4 (TRL-4) or higher, excluding early-stage ideas.
10. This TRL-4 condition is seen as arbitrary and contrary to the nature of innovation, which often requires funding at much earlier stages.
11. Unlike the venture capital model, which accepts risk and unpredictability, this scheme may favor only safe bets.
12. Advanced countries built technological strength through military-industrial investments, funding risky technologies that later benefited civilians.
13. India lacks such strategic, long-term investment frameworks, especially those that tolerate high-risk research.
14. A major issue remains the brain drain of Indian scientists, who move abroad due to the lack of domestic opportunities.
15. The editorial concludes that budgetary funding alone is insufficient and structural reforms are needed to truly transform India's R&D ecosystem.

Practice Exercise: SSC Pattern Based



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